KARTHIKEYA NAGISETTI

AI Engineer · Machine Learning Systems · LLM Fine-Tuning & RAG

· 25karthikeyanagisetti@gmail.com · +1 469-269-7704 · linkedin.com/in/karthikeya-nagisetti · github.com/25karthikeyanagisetti-code

SUMMARY

AI Engineer at Meta. 3+ years shipping production ML — LLM fine-tuning on 8× A100s, hybrid RAG over internal docs, fraud detection at ~600 req/sec at p99 < 120 ms. End-to-end: PyTorch training loops to Triton inference, MLOps to on-call. **Author of Snap-Time Agent** — an open-source Bounded-Autonomy framework for LLM agents (emotion-vector cognition, importance-weighted memory, live interactive demo).

TECHNICAL SKILLS

ML / DL: PyTorch, Hugging Face Transformers, TRL, LoRA / QLoRA, DeepSpeed (ZeRO-2/3), ONNX, scikit-learn, XGBoost
LLMs & RAG: LLaMA 2/3, Mistral, GPT-4 / Claude APIs, prompt engineering, vector retrieval (pgvector, FAISS), eval harnesses, guardrails
Languages: Python (expert), SQL, Bash, C++ (intermediate), TypeScript
Infrastructure: Docker, Kubernetes, AWS (S3, EC2, SageMaker), GCP (Vertex), NVIDIA Triton Inference Server, Linux
MLOps: MLflow, Weights & Biases, Optuna, Airflow, GitHub Actions, Terraform (basic), shadow A/B testing
Data: PostgreSQL (+ pgvector), Redis, Spark / PySpark, DuckDB, Pandas, NetworkX, DGL

EXPERIENCE

Meta — AI Engineer — Menlo Park, CA · Dec 2025 – Present

- Contributing to **Deployment Buddy**, an internal LLM-powered assistant on top of LLaMA 3.x, built on Meta's inference platform.
- Implemented a **hybrid retrieval pipeline** (vector + BM25) over internal docs feeding the assistant; cut average context-lookup time ~40% vs. the previous prototype.
- Built a **regression-eval harness** on a held-out 1.2K-prompt set; surfaced two recurring hallucination patterns that drove prompt-and-retrieval changes.
- Active in code review and on-call rotation; shipped 6 production CRs in the first quarter post-onboarding.

Healed Hearts — Lead AI Engineer — San Francisco, CA · Jun 2025 – Nov 2025

- Fine-tuned **Llama-2-13B and Mistral-7B** on a 200K-sample clinical-notes dataset using LoRA + 4-bit QLoRA on 8× A100s; reached **87% accuracy on a USMLE-Step-1-style multiple-choice eval** (vs. 81% base).
- Built a **clinical RAG pipeline** (Postgres + pgvector over PubMed abstracts) behind a FastAPI service handling ~5K daily requests at p95 ~280 ms.
- Added a Constitutional-AI-style critic loop and contributed to the eval framework that flagged hallucinated citations before release; ran initial DP-SGD experiments for a partner-data fine-tune.

U.S. Bank — AI Engineer — San Francisco Bay Area · Jun 2024 – May 2025

- Shipped a **fraud-detection model** combining XGBoost with graph features (NetworkX → DGL); **cut false positives ~22%** at the precision target (99.4%) on the production stream.
- Migrated a legacy NLP pipeline (regex + scikit-learn) to a fine-tuned **DistilBERT classifier** for financial-document categorization; **+9 F1 points** and ~3× faster inference after ONNX export.
- Stood up model serving on Kubernetes (Triton + a FastAPI gateway) handling ~600 req/sec at p99 < 120 ms; wrote MLflow + Airflow pipelines for nightly retraining and shadow A/B (release cycle: ~3 weeks → ~3 days).

Mivada — Machine Learning Engineer — Sydney, Australia · Jan 2024 – May 2024

- Implemented a multi-lingual classification head on top of XLM-RoBERTa covering 7 languages; +6 F1 over single-language baselines. Profiled the training loop and switched to fp16 + gradient accumulation, halving single-GPU training time without accuracy regression.

Ernst & Young LLP — AI Research Associate — Bengaluru, India · Feb 2023 – Jul 2023

- Built NLP pipelines (FinBERT, RoBERTa) for financial-filing analysis — ~91% F1 on document-section classification; contributed to a fraud-detection prototype (XGBoost + graph features), +12% recall over the existing baseline in offline evals; built an internal Optuna wrapper for hyperparameter sweeps used by 4 team projects.

INDEPENDENT RESEARCH & PROJECTS

- **Snap-Time Agent — Human-Like AI** (independent research) — Repo + live interactive demo for a Bounded-Autonomy framework: LLM agents with multidimensional emotion vectors, importance-weighted temporal memory, and a Snap-Time decision protocol for behavior under deadline pressure; unified loss adds emotion-regulation + memory-decay terms. **github.com/25karthikeyanagisetti-code/snap-time-agent** · **25karthikeyanagisetti-code.github.io/snap-time-agent**
- **Open-source & writeups** — Notes on RAG retrieval / hallucination-eval patterns; small PRs to Hugging Face Transformers (docs + tokenizer edge case).

EDUCATION

M.S., Technology Management, Lindsey Wilson College · GPA 3.8 — Columbia, KY · 2023 – 2025
B.S., Electrical Engineering, Vardhaman College of Engineering · GPA 3.2 — Hyderabad, India · 2019 – 2023

Certifications: AWS Machine Learning – Specialty · Google Professional ML Engineer